University course registration system

Contents

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0: Problem Statement

Rajarata University currently manages the module registration process via physical forms or email. This causes Slow processing time – Staff must manually review and enter information, Human errors – Forms can be filled incorrectly, or data can be entered inaccurately, Lost or misplaced forms/emails – Paper forms can be misplaced. Emails may be overlooked or end up in spam, Duplicate registrations – Students may submit multiple forms or emails by mistake, Lack of real-time updates – Students cannot instantly see whether their registration has been approved or if modules are full, Delays in communication – Back-and-forth emails to correct mistakes can slow the process.

1: Stakeholder Analysis & RACI Matrix

1. a: Stakeholder Analysis

Manager

Receptionist

Lecturers

Students

Staff

Developers

BA

1.b: Power/Interest Grid

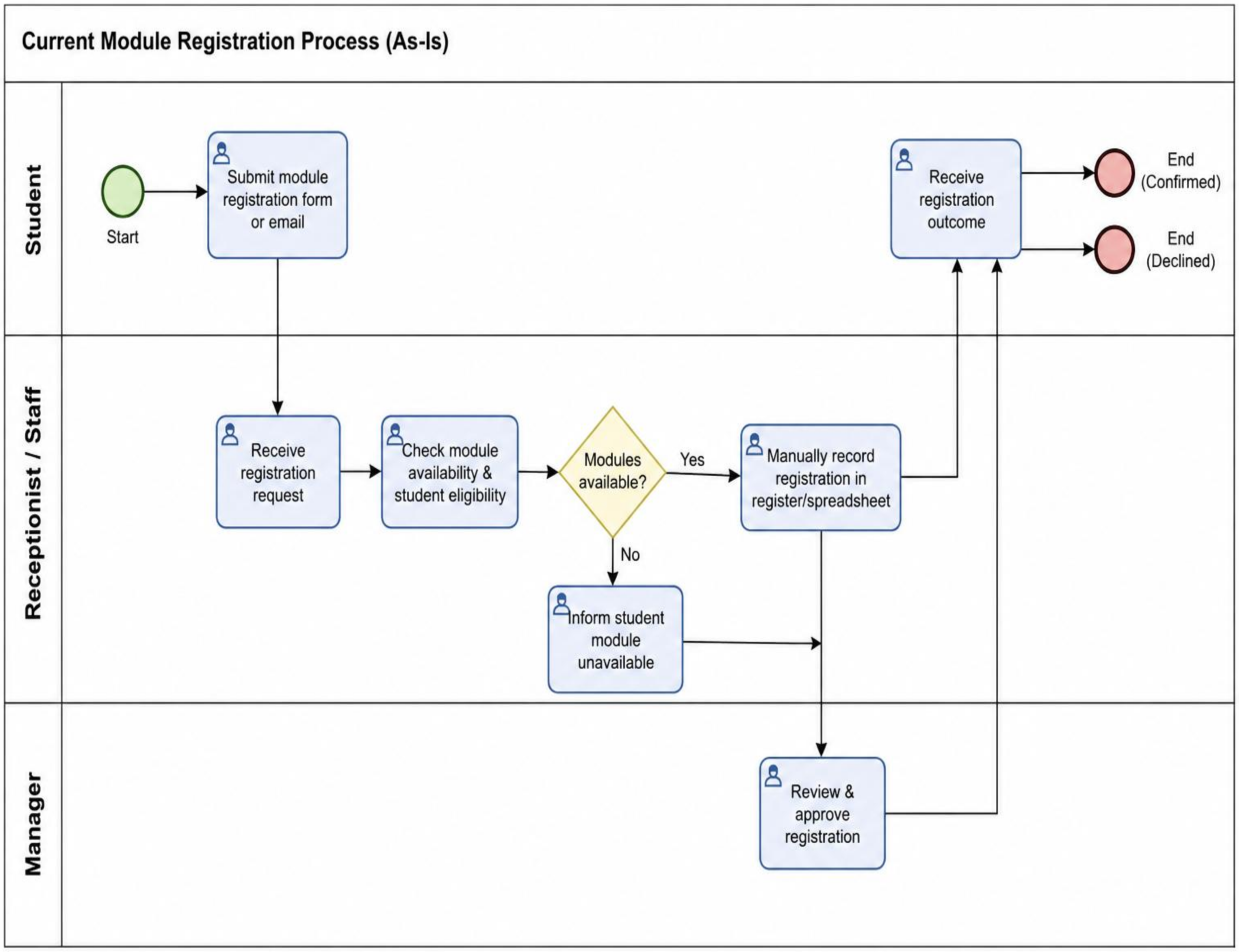
Stakeholder	**Power**	**Interest**	**Engagement Strategy**
Manager	High	High	Manage closely - Weekly updates
Receptionist	Low	High	Keep informed
Lecturers	Low	Medium	Keep informed
Staff	Low	Medium	Keep informed

Students	Low	High	Gather feedback via survey/UAT
Developer	High	High	Manage closely

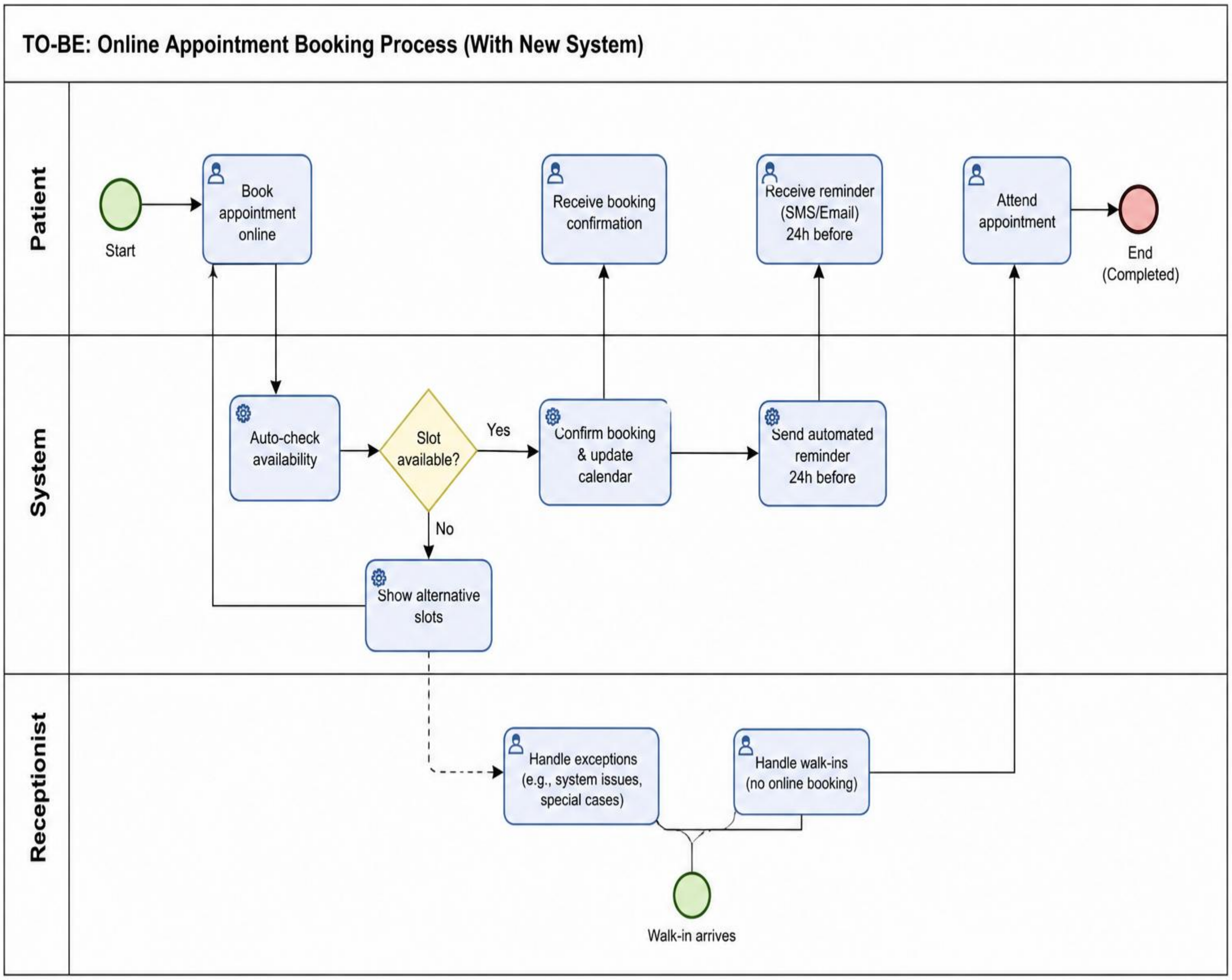
1.c: RACI matrix

Activity	Manager	Receptionist	Developer	BA
Define requirements	A	C	I	R
Approve BRD	A	I	I	R
Build System	I	I	R	C
UAT sign-off	A	R	I	C

2:AS-IS Process Map



3:TO-BE Process Map



4: SRS Document

1. Introduction

1.1 Problem Statement

The current university module registration process is handled manually through physical forms or email communication. This causes delays, miscommunication, and lack of real-time visibility of available module slots. Students often face uncertainty regarding registration status, while staff must manually verify availability and process approvals, increasing workload and errors.

1.2 Proposed Solution

A streamlined digital module registration system that allows students to submit requests online, automatically checks module availability, and routes approvals through academic staff for faster processing and improved accuracy.

2. Scope

2.1 In Scope

- Student online module registration submission
- Automatic module availability checking
- Staff/receptionist review and processing
- Manager/Academic officer approval workflow
- Confirmation notification to students (email/SMS optional if included in your model)

2.2 Out of Scope

- Online payment processing
- Advanced AI-based course recommendations
- Real-time chat between students and staff
- External university system integrations

3. Stakeholders

- **Student** – initiates registration, views status, receives confirmation
- **Receptionist / Registration Staff** – processes requests and validates data
- **Manager** – final approval authority

- **University System Admin**– maintains system configuration and user access

4. Business Requirements (High-Level Goals)

ID	Requirement
BR-01	Reduce module registration processing time by at least 50%
BR-02	Minimize manual errors in module allocation
BR-03	Improve transparency of registration status for students
BR-04	Ensure accurate and real-time module availability tracking
BR-05	Reduce administrative workload on staff

5. Functional Requirements

ID	Requirement	Priority
FR-01	Students can submit module registration requests online	Must
FR-02	System validates module availability automatically	Must
FR-03	System displays confirmation or rejection status to student	Must
FR-04	Staff can view all incoming registration requests	Must
FR-05	Staff can approve or reject registration requests	Must
FR-06	Manager can give final approval	Must
FR-07	System logs all registration actions for auditing	Should
FR-08	Students receive notification upon status update	Could
FR-09	System supports multiple departments or faculties	Won’t

6. Non-Functional Requirements

- **NFR-01:** System must support at least 500 concurrent student requests during peak registration periods
- **NFR-02:** System response time must be under 3 seconds for all main actions
- **NFR-03:** System must ensure secure authentication for all users
- **NFR-04:** Data must be stored securely and comply with university data protection policies
- **NFR-05:** System should be available 99% of the time during registration periods

7. Assumptions & Constraints

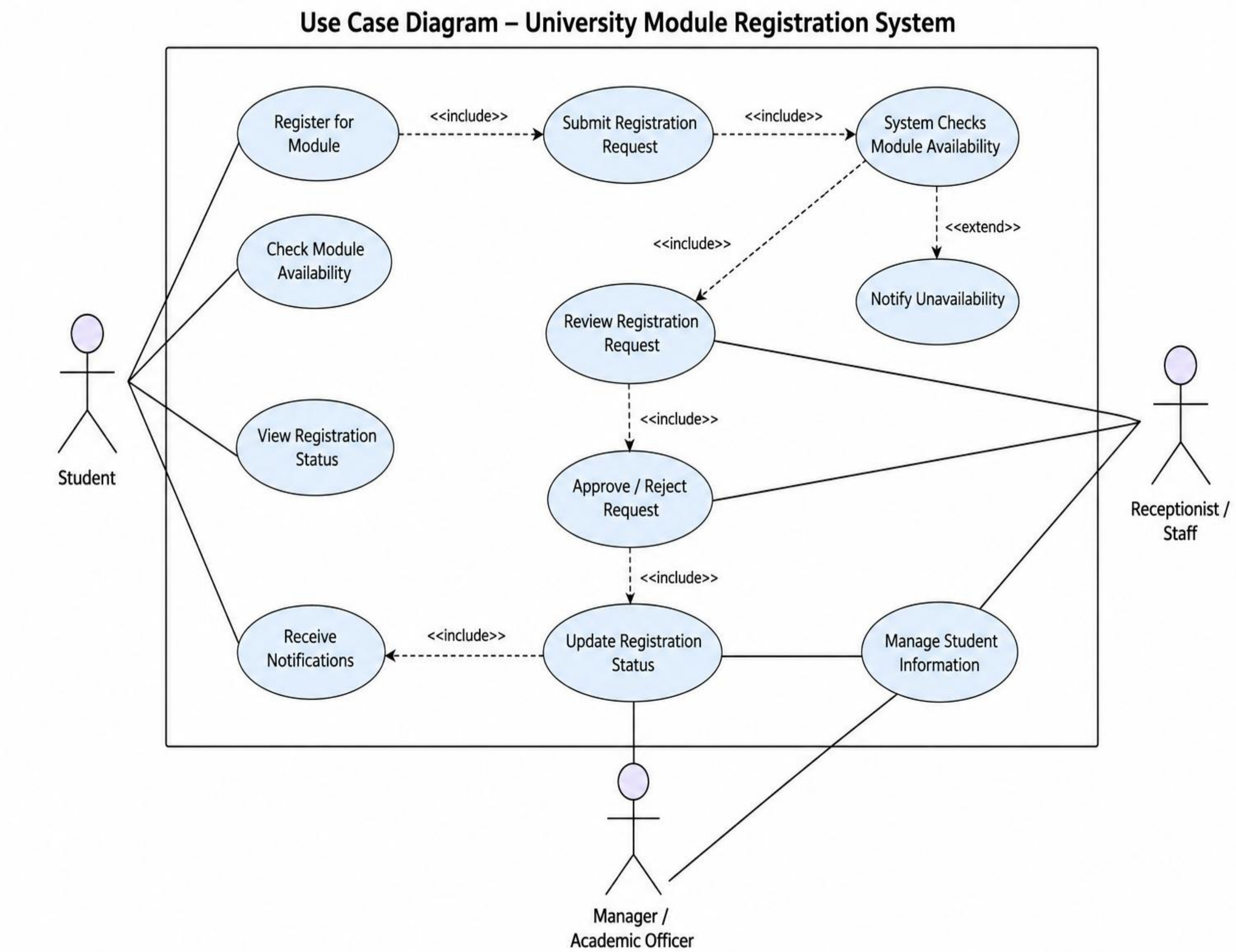
Assumptions

- Students already have valid university login credentials
- Module availability data is maintained by academic staff
- Staff are trained to use the system interface

Constraints

- No integration with external payment gateways in Version 1
- System depends on accurate manual entry of module data by staff
- Internet access is required for all users

8. Use Case Summary



5: Requirements Traceability Matrix

ID	**Description**	**Source**	**Test Case**	**Status**
FR-01	Students can submit module registration online	BR-01	TC-01	Planned
FR-02	System checks module availability automatically	BR-01	TC-02	Planned
FR-03	System shows confirmation/rejection status	BR-03	TC-03	Planned
FR-04	Staff can view registration requests	BR-05	TC-04	Planned
FR-05	Staff can approve/reject requests	BR-05	TC-05	Planned
FR-06	Manager gives final approval	BR-03	TC-06	Planned

6: Business Case-Digital Registration System

1. Problem

The current university module registration process is manual, relying on physical forms and email communication. This leads to delays in processing, frequent errors in module allocation, and poor visibility for students regarding their registration status. Administrative staff also experience high workload during peak registration periods.

2. Proposed Solution

A web-based module registration system that allows students to submit requests online, automatically checks module availability, and routes approvals through academic staff. The system provides real-time status updates and reduces manual handling of registration requests.

3. Estimated Cost (Year 1)

Assumptions:

- Freelance developer rate: LKR 2,500/hour
- Estimated development time: 120 hours
- Hosting & maintenance: LKR 8,000/month

Cost Breakdown:

- Development cost: 120 × 2,500 = **LKR 300,000**
- Hosting (Year 1): 8,000 × 12 = **LKR 96,000**

Total Year 1 Cost ≈ LKR 396,000

4. Estimated Benefit

Assumptions:

- 500 students register per semester
- Peak load: ~40 registration actions per day
- Manual process causes ~10% delays/errors currently
- System reduces inefficiency by ~50%

Estimated impact:

- Prevents ~4–5 registration errors/delays per day
- Saves ~30 minutes of staff time per error case

- Total saved staff time ≈ 2–3 hours/day

Monetized benefit (staff time value assumption):

- Staff cost equivalent: LKR 1,500/hour
- Daily savings: ~2.5 hours → LKR 3,750/day
- Annual academic days (~200 days):
  → **LKR 750,000/year saved**

5. Simple ROI

- Total Cost (Year 1): LKR 396,000
- Annual Benefit: LKR 750,000

Net Benefit: 750,000 − 396,000 = **LKR 354,000**

ROI: (354,000 / 396,000) × 100 ≈ **89% ROI (Year 1)**
Payback Period: ~6–7 months

6. Recommendation

It is recommended to proceed with the development of the digital module registration system. The solution provides strong cost savings, reduces administrative workload, and significantly improves student experience. A phased rollout (starting with student registration and approval workflow) is advised to minimize implementation risk.